Problem 1. Least Squares and Minimum-Norm Solutions (a) Overdetermined least squares via SVD (6 points). Let $A \in \mathbb{R}^{m\times n}$, $m > n$, have full column rank, and let $b \in \mathbb{R}^m$. Consider

$$\min_{x\in\mathbb{R}^n} \|Ax - b\|_2.$$

Suppose a thin SVD of A is given by

$$A = U\Sigma V^T,$$

where $U \in \mathbb{R}^{m\times n}$ has orthonormal columns, $\Sigma \in \mathbb{R}^{n\times n}$ is diagonal with positive diagonal entries, and $V \in \mathbb{R}^{n\times n}$ is orthogonal.

(i) Derive an expression for the unique least-squares solution x_* in terms of U, Σ, V, and b. Prove that your expression minimizes $\|Ax - b\|_2$.

(ii) Show that x_* satisfies the normal equations $A^TAx_* = A^Tb$.

(iii) Express the minimum residual norm $\|Ax_* - b\|_2$ in terms of U and b.

(b) Underdetermined minimum-norm solutions via SVD (8 points). Now let $A \in \mathbb{R}^{m\times n}$, $m < n$, have full row rank, and let $b \in \mathbb{R}^m$. Consider

$$\min_{x\in\mathbb{R}^n} \|x\|_2 \qquad \text{subject to } Ax = b.$$

Suppose a thin SVD of A is given by $A = U\Sigma V^T$, where $U \in \mathbb{R}^{m\times m}$ is orthogonal, $\Sigma \in \mathbb{R}^{m\times m}$ is diagonal with positive diagonal entries, and $V \in \mathbb{R}^{n\times m}$ has orthonormal columns.

(i) Explain why $Ax = b$ is consistent for every $b \in \mathbb{R}^m$.

(ii) Derive an expression for the unique minimum-norm solution x_* in terms of the SVD factors. Prove both feasibility and minimum norm.

(iii) Show that $x_* = A^T(AA^T)^{-1}b$.

(c) Minimum-norm solutions via QR (8 points). Continue with the assumptions in part (b). Suppose a reduced QR factorization of A^T is given by

$$A^T = QR,$$

where $Q \in \mathbb{R}^{n\times m}$ has orthonormal columns and $R \in \mathbb{R}^{m\times m}$ is nonsingular and upper triangular.

(i) Show that the minimum-norm solution is $x_* = QR^{-T}b$.

(ii) Describe an algorithm to compute x_* using Q, R, and b, without explicitly computing any inverse.

(iii) Explain why this algorithm gives the same solution as in part (b).

(d) Geometry and normal equations (3 points). Continue with the assumptions in part (b).

(i) Explain why the normal equations $A^TAx = A^Tb$ do not determine a unique vector x.

(ii) Show that the minimum-norm solution is the unique solution of $Ax = b$ belonging to range(A^T). Explain briefly how the columns of Q in part (c) relate to this subspace.

Problem 2. Cholesky QR. Let $A \in \mathbb{R}^{m \times n}$, with $m \geq n$, have full column rank.

Consider the following algorithm for computing a QR factorization of A:

$$\begin{aligned} G &= A^T A, \\ R &= \text{chol}(G), \\ Q &= AR^{-1}, \end{aligned}$$

where $\text{chol}(G)$ denotes the upper triangular Cholesky factor of G, satisfying $G = R^T R$.

(a) Prove that the algorithm is well-defined, i.e., that the Cholesky factorization exists and that R is nonsingular.

(b) Prove that the matrices Q and R computed by the algorithm constitute a reduced QR factorization of A.

(c) Explain how the computation $Q = AR^{-1}$ can be implemented without explicitly forming R^{-1}.